

Modern Application Development 1 Project Report

1.Student Details

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About Me: I am a student in the Diploma level of the IITM BS Degree.

2.Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build an efficient to manage campus recruitment activities involving companies, students, and placement drives.

Approach:

The app was built using Flask as the backend framework. It allows companies to create placement drives and manage applications and students to create their profiles, apply for placement drives and track their application status.

3.AI/LLM Declaration

I used ChatGPT (GPT-5) to assist in writing SQLAlchemy model definitions and a tiny percentage of it in improving and making certain SQLAlchemy queries more efficient. It has also been used in improving the looks of some pages(Bootstrap).

The extent of AI/LLM usage is around 15%, limited to code suggestions and frontend formatting.All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
Flask-WTF	Forms
WTForms	Frontend form validation
SQLite	Lightweight local database for storing user data

5. Architecture and Features

Architecture Overview:

- plc.py – main Flask application entry point
- app – package that contains the routes , model ,forms and templates
- model.py – database models using SQLAlchemy
- student.py – Flask Blueprint for student routes
- company.py – Flask Blueprint for company routes
- admin.py – Flask Blueprint for admin routes
- miscellaneous.py – Flask Blueprint for common routes
- form.py – Flask-WTF form classes
- /templates – Jinja2 HTML templates
- /static – for storing resumes

Implemented Features:

Company and student registration and login, admin login

Drive creation by companies and managing applications

Students have their own profiles and apply to drives and check the progress

Admin can approve companies and drives and blacklist users

Secure password hashing and user session handling is implemented

6.Database Schema/ER Diagram

Tables:

User - stores authentication and basic account details (id, username, name, password_hash, role, is_blacklisted, created_at)

Student - stores student-specific profile details (id, user_id, roll_number, department, cgpa, graduation_year, resume)

Company - stores company profile details (id, user_id, company_name, hr_contact, website, approval_status)

PlacementDrive - stores job/internship opportunities created by companies (id, company_id, job_title, job_description, eligibility_criteria, application_deadline, status, created_at)

Application - stores student applications to placement drives (id, student_id, drive_id, applied_at, status)

Constraints:

Composite Unique Constraint → (student_id, drive_id) in Application table to prevent duplicate applications.

Relationships:

One-to-One → User → Student

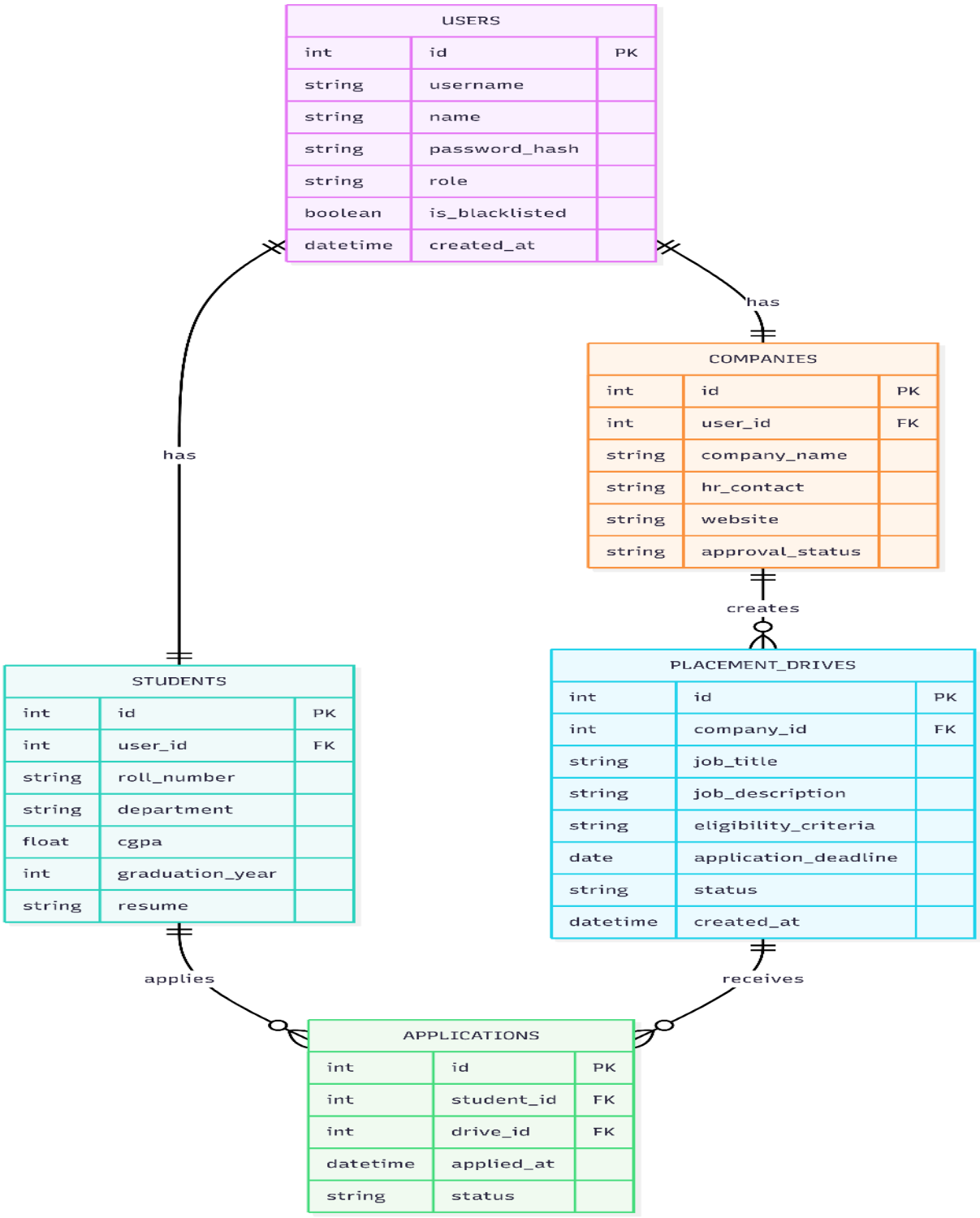
One-to-One → User → Company

One-to-Many → Company → PlacementDrive

One-to-Many → Student → Application

One-to-Many → PlacementDrive → Application

ER diagram below:



7.Video Presentation

<https://drive.google.com/file/d/1sm36YTHRia0szUru6rW1aQfZfrj-2b7l/view?usp=sharing>